

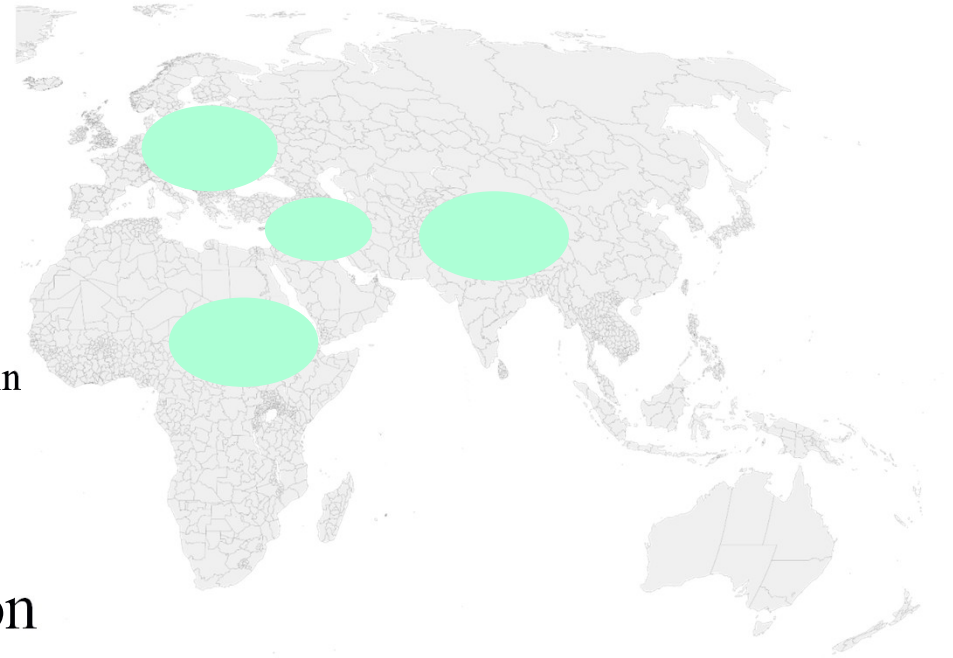


GLOBAL SHEEPPOX POPULATION STRUCTURE BASED ON WHOLE GENOME SEQUENCES

Floris Breman

Sheeppox what we (do not) know

- Origins unknown. Asia, Europe, Africa?
- Known from historical sources in Europe and Asia (>1000y ago) (Hutyra 1912)
 - Availability of historical literature describing pox like symptoms is limited, but exists (Babkin et al. 2015)
- Rise and dispersal of SPPV (and GTPV) may have coincided with the domestication of sheep (and goats)
 - Similar to the rise of *Variola*, which is associated with human population increase and urbanization (Babkin et al. 2015)?



Sheeppox what we (do not) know

- Mortality ranges from almost 0% to almost 100% (Tuppurainen et al. 2015)
- European sheep breeds seem to be particularly vulnerable (Davies 1991, Bhanuprakash et al. 2006)
- Approximately 37 whole genome sequences available in public database as of September 2025 and only ~700 other sequences (from 60-4000bp)

Explore Virus Data

Applied Filters: Virus/Taxonomy (1) ✕

Selected Results: 0 / 784

Align Build Phylogenetic Tree Download All Results

Nucleotide (784) Protein (5,437) NCBI Virus Assembly (1) i

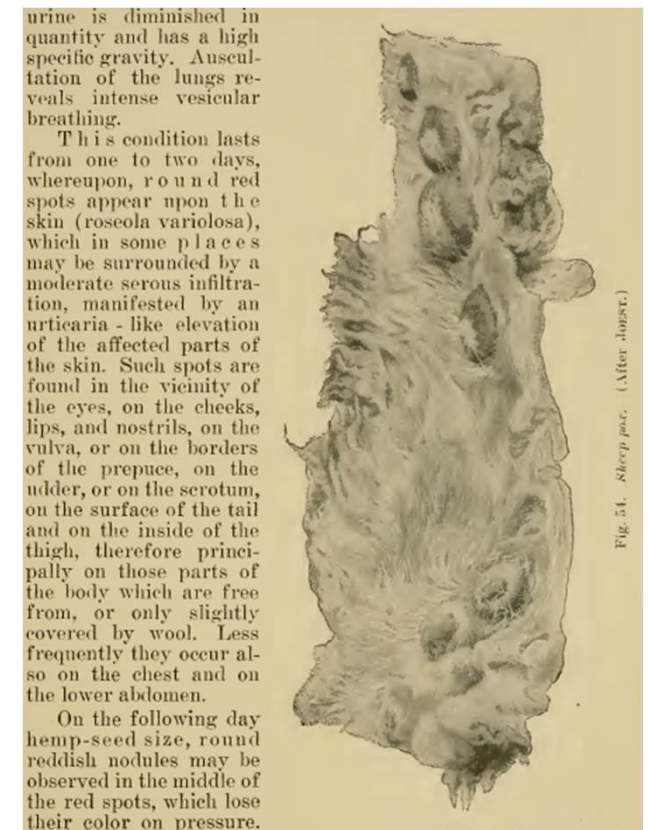
Accession Organism Name Species

Species	Genotype	Isolate	Length
Capripoxvirus sheeppox		V330	150491
Capripoxvirus sheeppox		V293	150478
Capripoxvirus sheeppox		V123	150369
Capripoxvirus sheeppox		V104	150211
Capripoxvirus sheeppox		SPPV-GH	150200

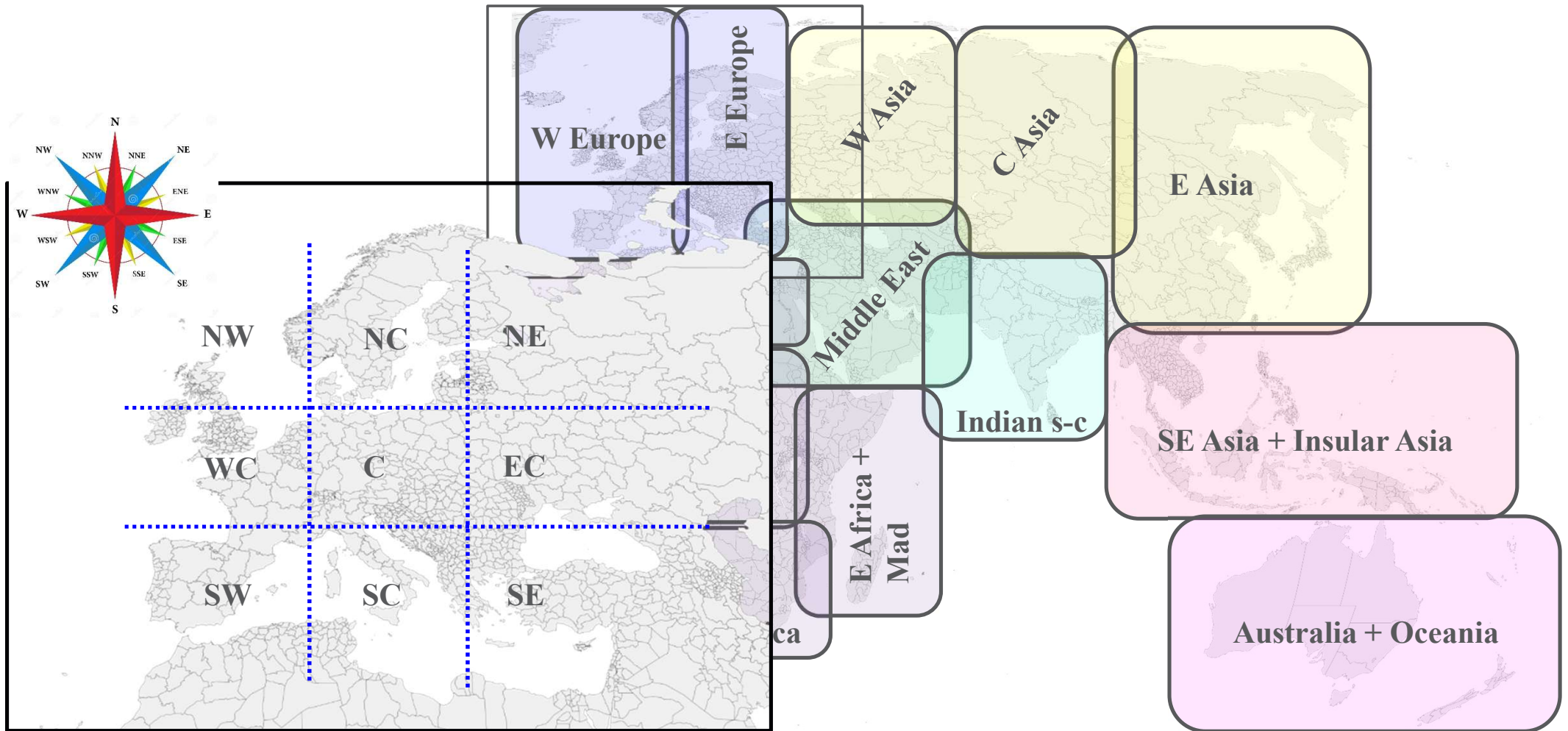


SPPV historical context

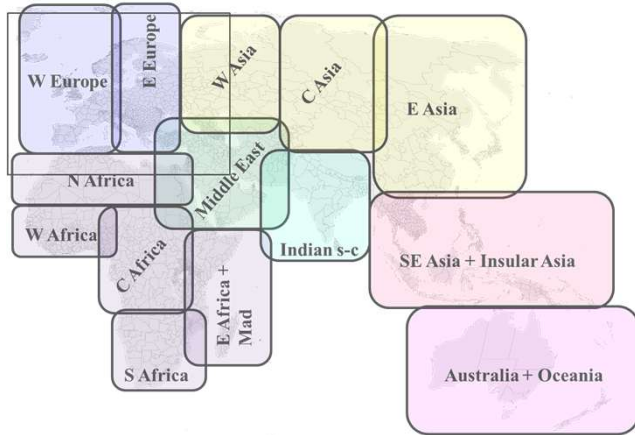
- Occurred widely in Europe
- Based on a review by Hutyra and Marek (*Variola ovina*) from 1912
 - Based on literature from the 19th century
- Experiments performed as early as 1763
- Earlier ‘anecdotal’ reports from C-Asia, India and Europe report ‘pox’ in Sheep
 - Extent and precise nature unknown, no illustrations available, descriptions not translated



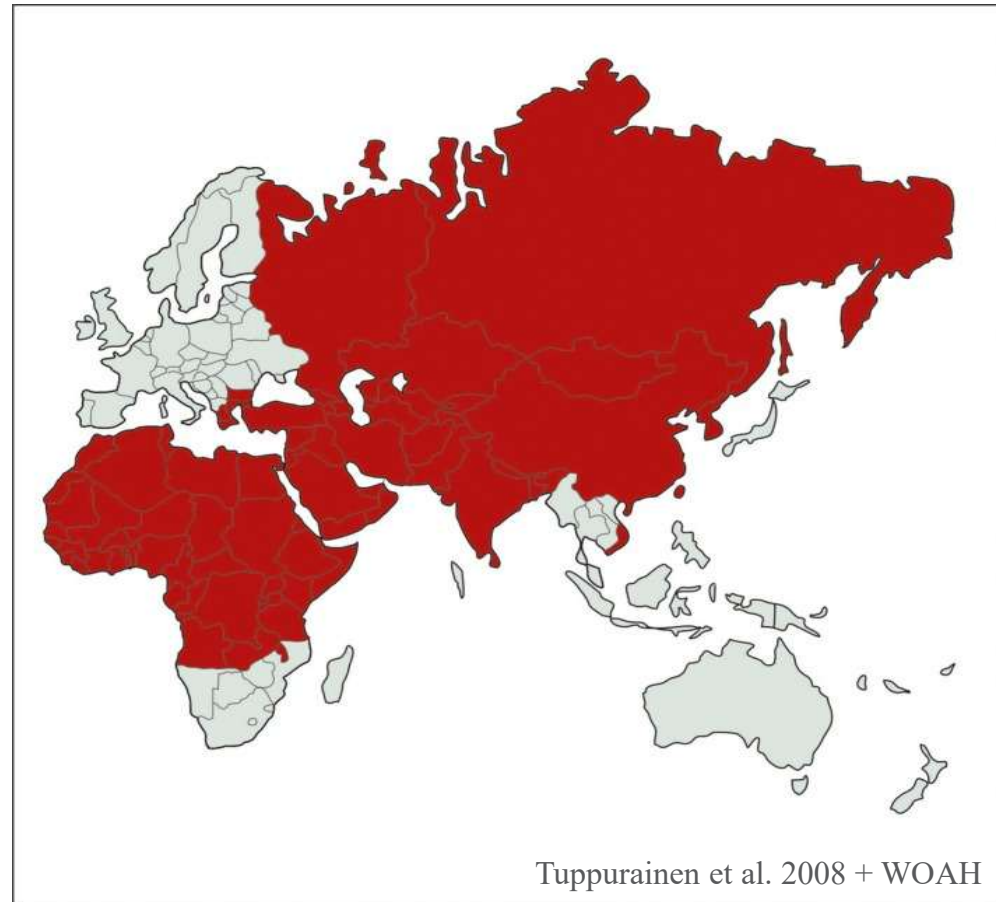
Practical division of global regions



Sheeppox (re)occurences



- West, E & C Asia
- The middle East
- The Indian Subcontinent
- N-Africa
- West, E & C Africa
- Repeated introductions in SE Europe



Two pre 1st world war outbreaks

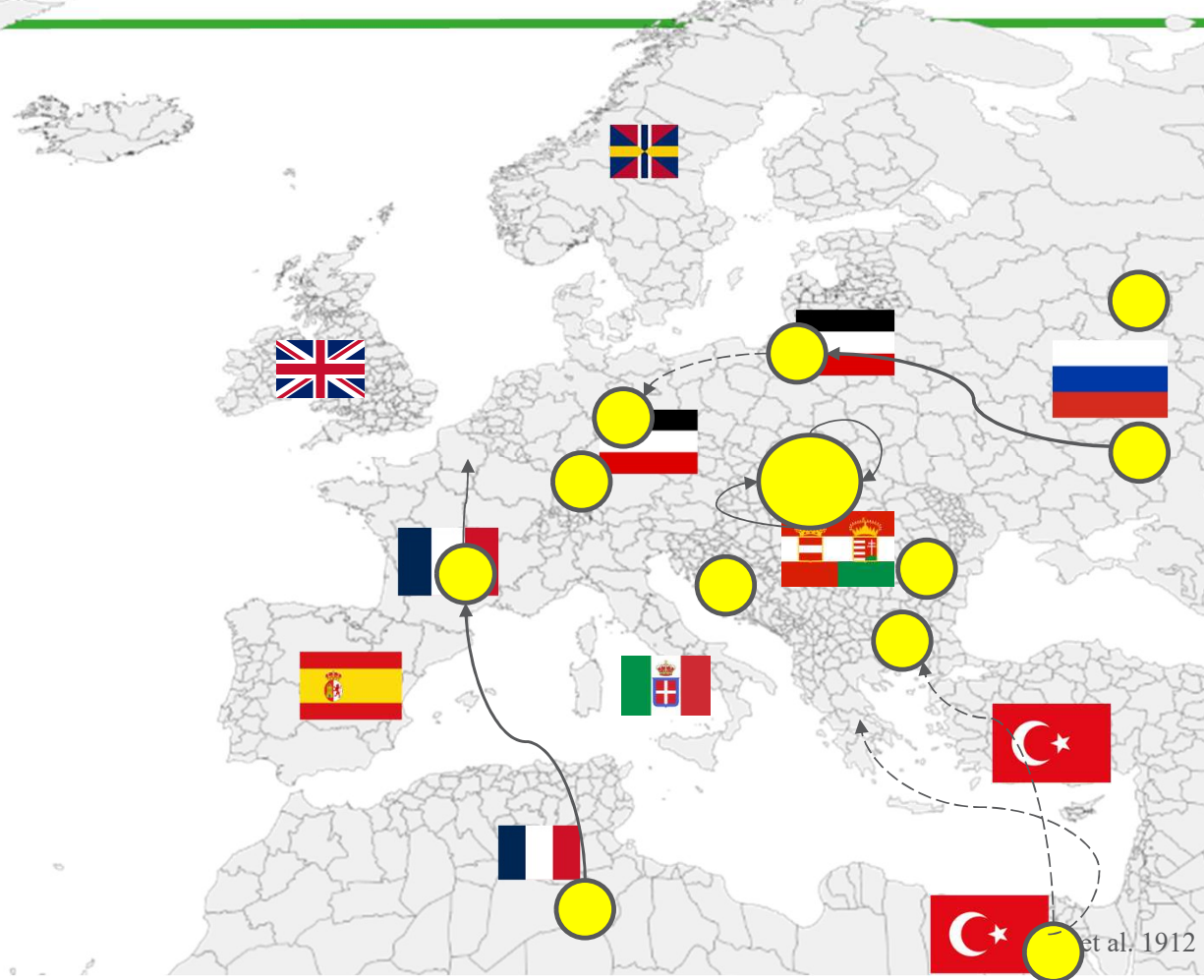


- Example from ~pre 1850
Estimated mortality >1.500.000
- Example from ~1890~1910
Estimated mortality ~50.000

- Vaccination under development
- quarantine and culling/‘stamping out’ were employed and seemed to have worked
- Transport routes were different then



Two pre 1st world war outbreaks

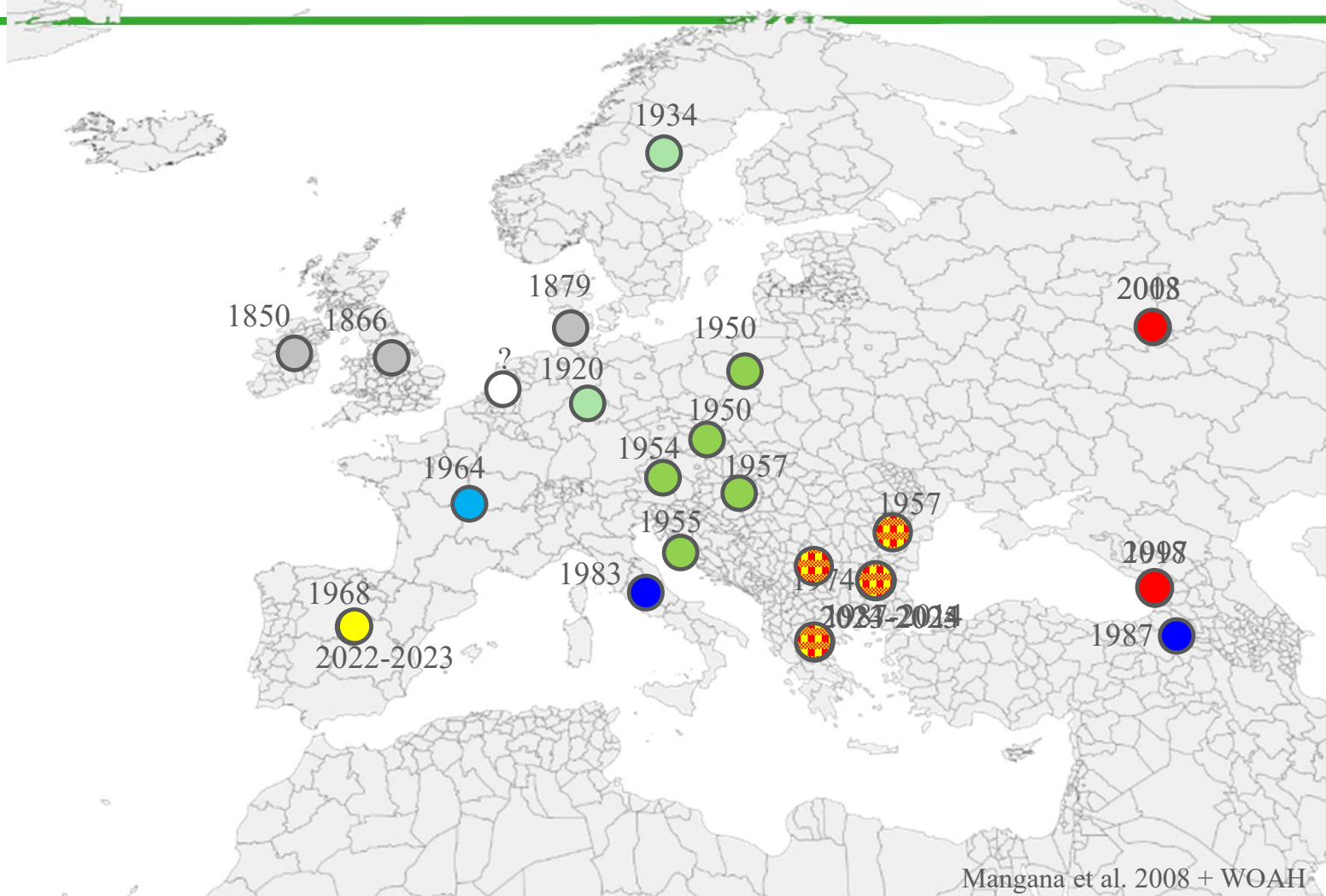


- Example from ~pre 1850
Estimated mortality >1.500.000
- Example from ~1890~1910
Estimated mortality ~50.000

- Vaccination under development
- quarantine and culling/‘stamping out’ were employed and seemed to have worked
- Transport routes were different then



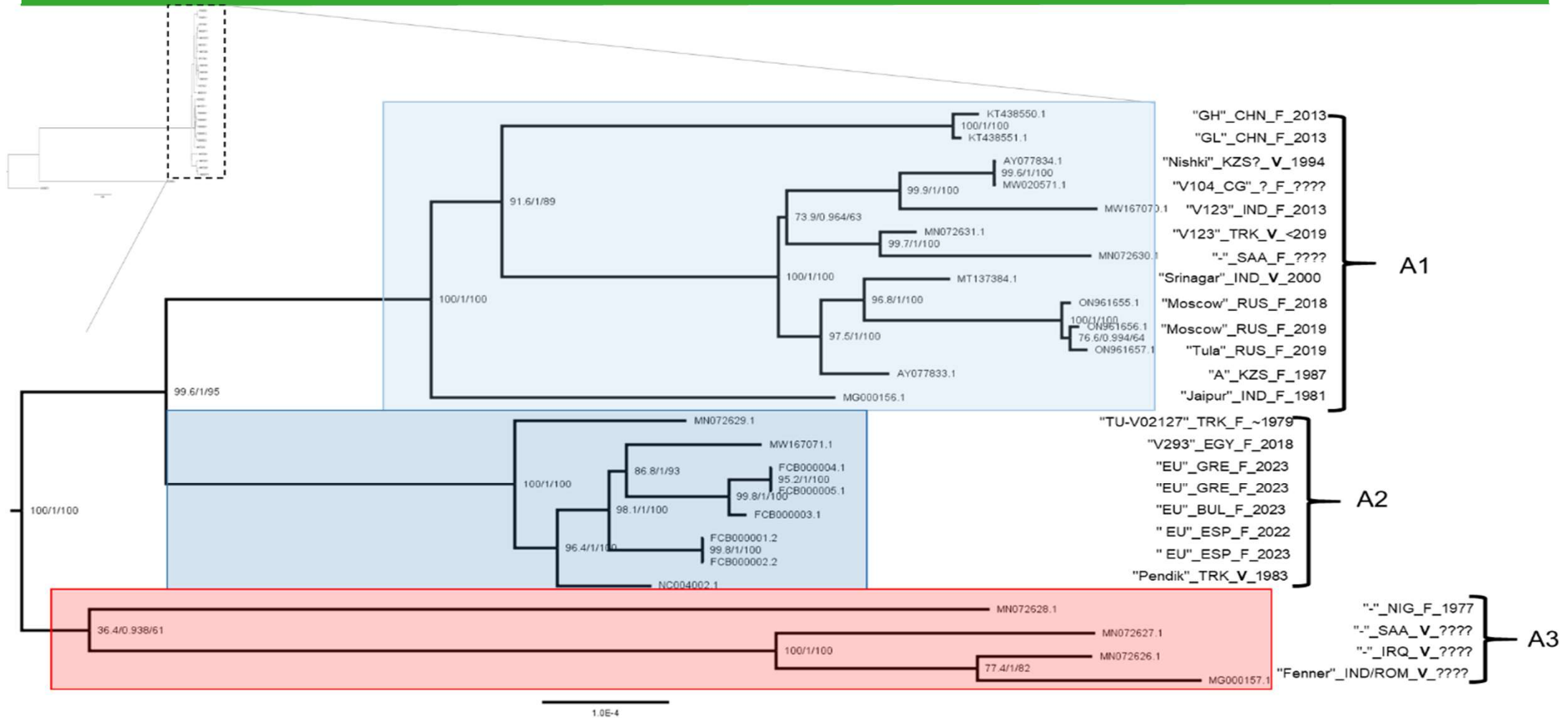
Final historical occurrences and date of eradication or reappearance



Mangana et al. 2008 + WOAH

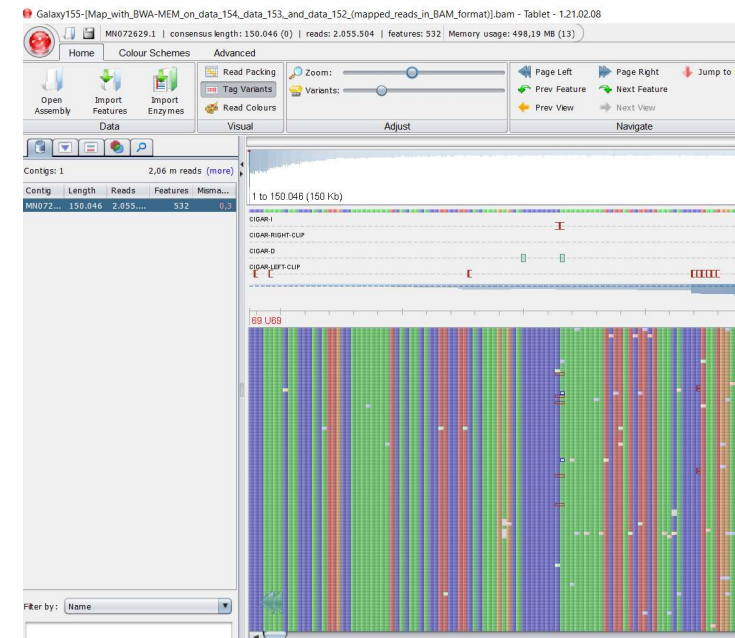
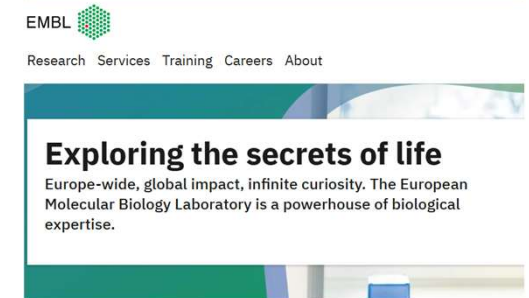


SPPV phylogeny 2024



New sequences

- From publicly available databases
- From Samples shared with the EURL/WOAH RL
 - sequenced using new insight gained from the past projects
- Regular DNA extraction
- NGS sequencing sensu (Breman et al submitted)
 - NGS using Illumina DNA prep
- Mapping using BWA and a reference
 - Iterative mapping if needed
 - Result manually evaluated

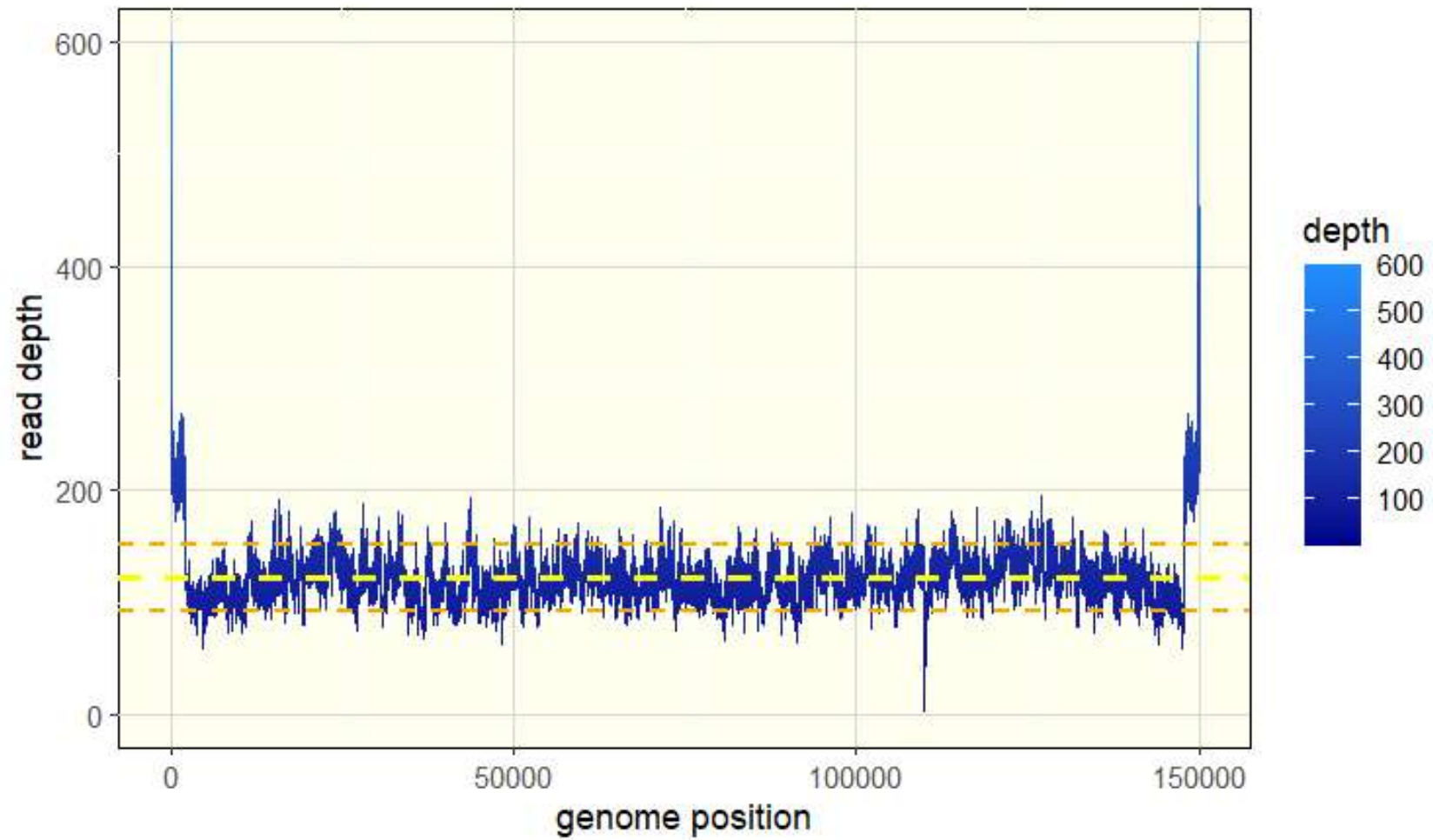


Sequencing results summary

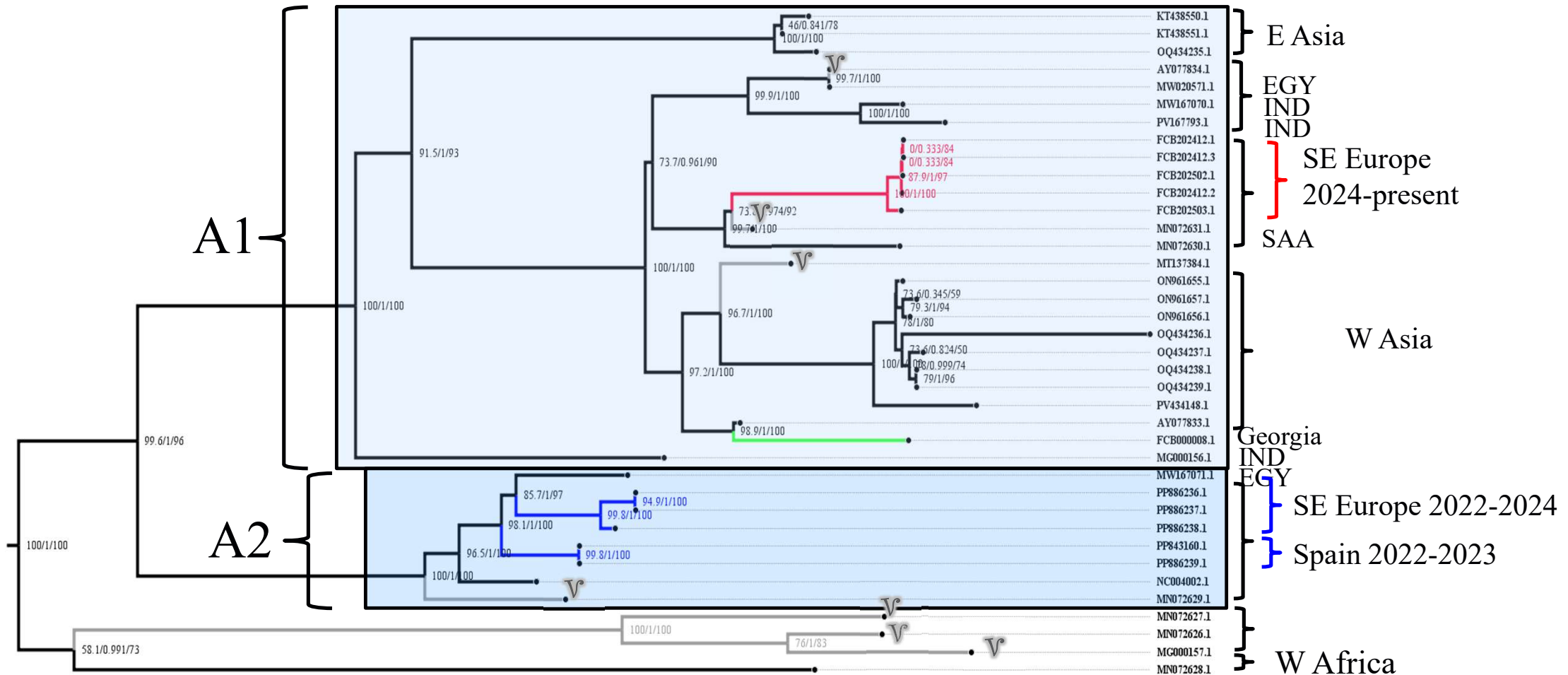
Genbank acc. Code	sample	total read count (*10 ⁶)	reads mapped (*10 ⁶)	percentage	CT-target D5R (Capripox)	CT non-target (beta actine)	ratio virus/host
NY	Bulgaria May 2025	2,7	0,12	4,44	11,87	17,96	0,66
NY	Bulgaria Aug 2025	1,38	0,027	1,98	13,56	19,36	0,70
NY	Romania 2025	10,8	0,037	0,34	13,47	18,94	0,71
PP843160	Andalucia (Granada)	53,4	2,7	5,1	17,17	24,48	0,70
PP886239	Castille La Mancha (Cuenca)	3,2	0,021	0,7	14,52	18,74	0,77
PP886238	Gorno Yabalkovo, Sredets (Burghas)	4,38	0,073	1,7	13,65	18,95	0,72
PP886236	Fthiotida	4,2	0,1	2,4	13,24	19,46	0,68
PP886237	Lesvos	4	0,08	2,0	14,12	19,81	0,71



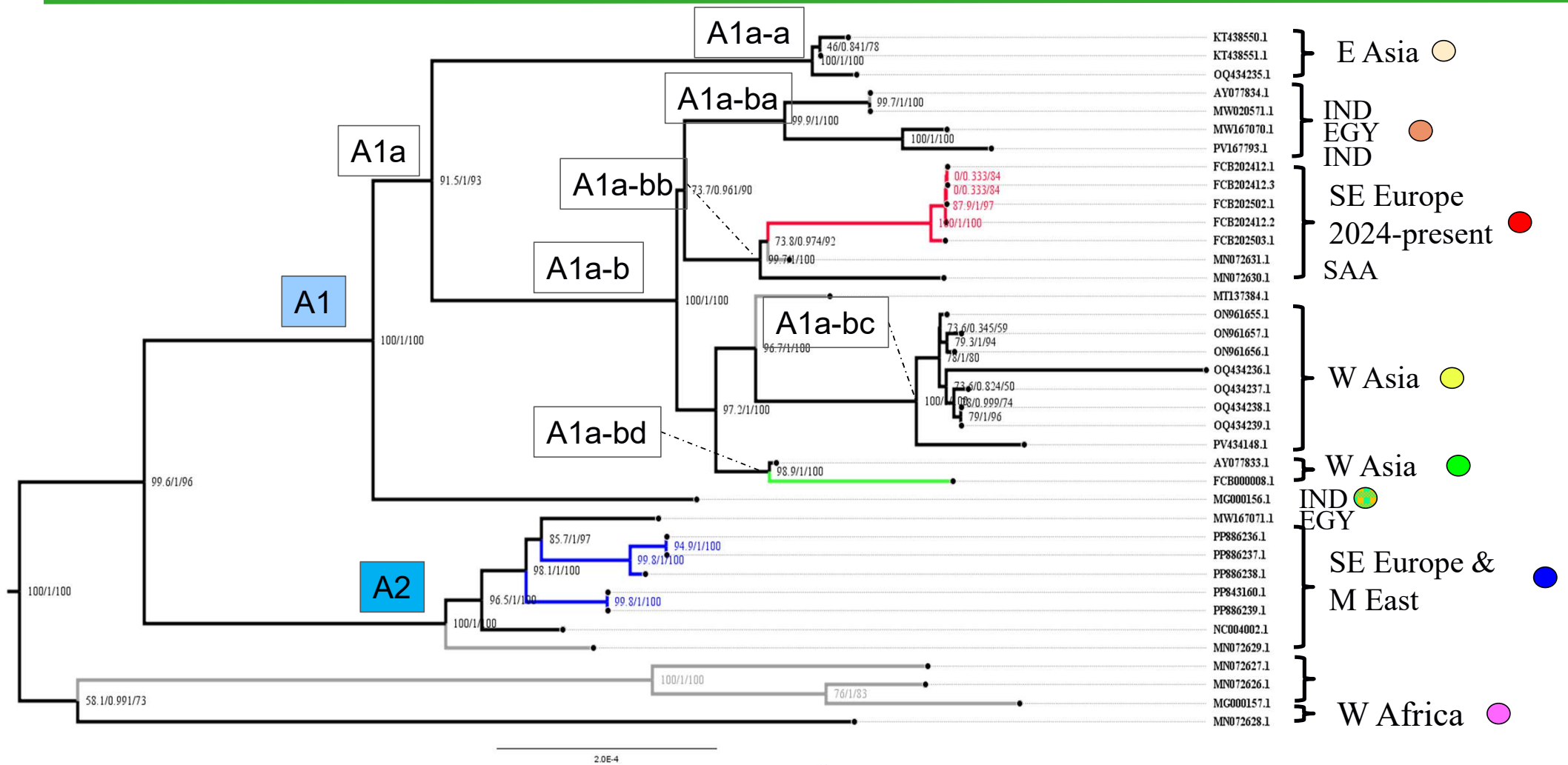
Sequencing results summary



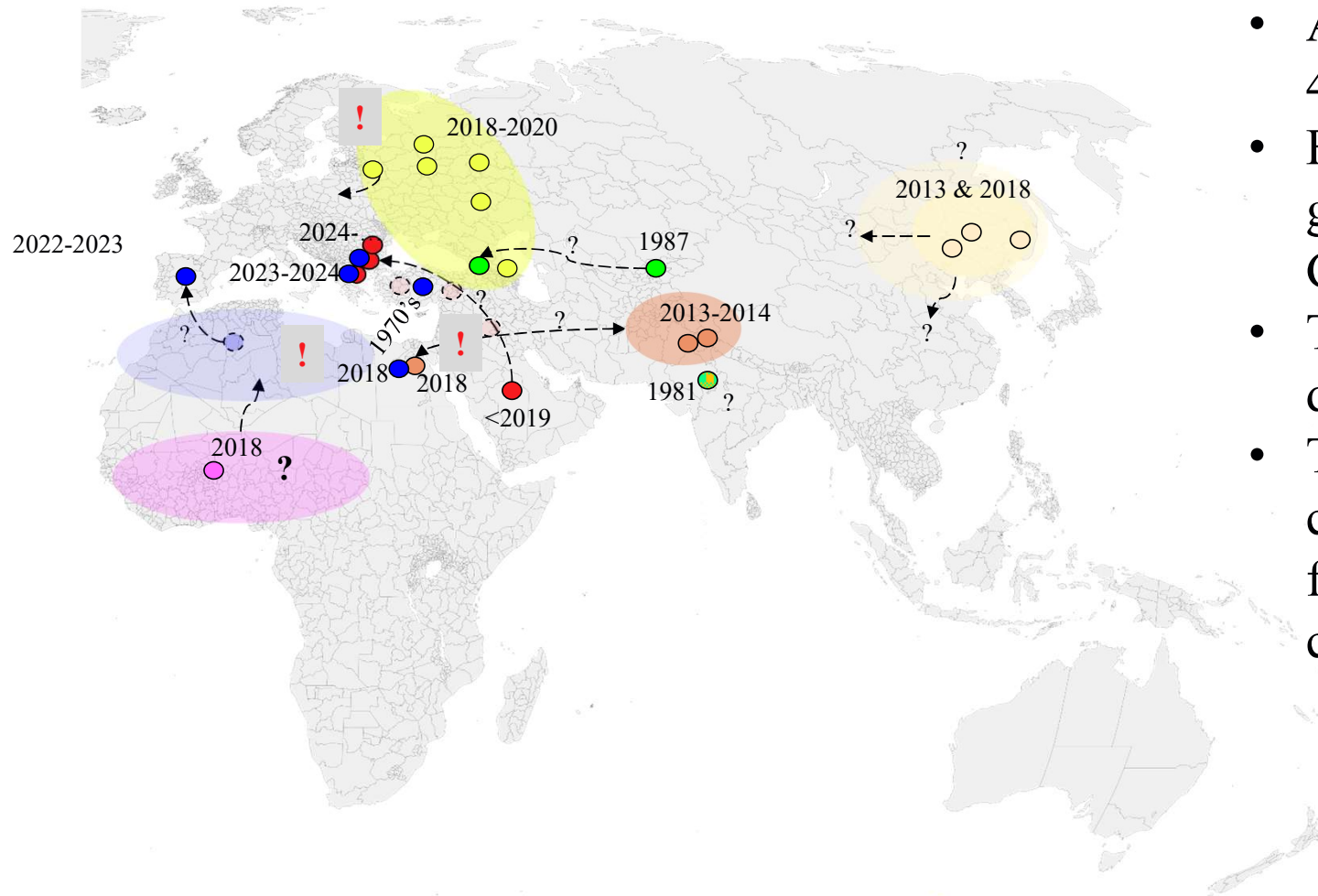
Updated phylogeny



Geographical distribution of clades/populations



Global distribution and gaps



- About 700 sequences (60-4000bp) available
- For only 25 countries there is genetic data (in any form) in GenBank
- This data is very unevenly distributed in time and space
- Take a look at the FMD community for a much closer follow up of dispersal and circulation



Discussion and reflection

- Dispersal routes, especially globally, are opaque
- More strains could enter Europe
 - Western Asian strains, currently circulating in Russia are a prime candidate
- Do different populations have different pathogenic or dispersal phenotypes?
- Vaccines seem to have been developed ad-hoc in a response to an outbreak, why?
 - Poor availability/expensive?
 - Poor quality/ineffective?
- There are some large gaps in geographical coverage of circulating SPPV strains
 - Northern Africa
 - Africa in general
 - Central & SE Asia
- Do we have historical samples?



Take home messages

- Past measures have worked and continue to be effective
- Continued vigilance is required
- Phylogenetics and geographical distribution of SPPV remain incomplete, but more sequences become available
- Collaboration is essential
- Detailed information is essential
- Historical context will be helpful



Acknowledgements

- Crew at the ExoVec laboratory, scientists and technicians
- The team at the NGS facility from Sciensano (TAG)
- The colleagues from Bulgaria, Greece, Romania and Georgia
- Numerous field workers and veterinarians
- European Commission for support



Thank you for your attention





Contact

Floris Breman • floris.Breman@sciensano.be • +32 23 79 06 06

Sciensano • Rue Juliette Wytsmanstraat 14 • 1050 Brussels • Belgium
T +32 2 642 51 11 • T Press +32 2 642 54 20 • info@sciensano.be • www.sciensano.be

